

## SAFETY DATA SHEET

Creation Date 15-Feb-2010

Revision Date 29-Mar-2024

Revision Number 3

### 1. Identification

|                             |                                                |
|-----------------------------|------------------------------------------------|
| <b>Product Name</b>         | <b>Potassium thiocyanate</b>                   |
| <b>Cat No. :</b>            | <b>14318</b>                                   |
| <b>CAS No</b>               | 333-20-0                                       |
| <b>Synonyms</b>             | Potassium rhodanide                            |
| <b>Recommended Use</b>      | Laboratory chemicals.                          |
| <b>Uses advised against</b> | Food, drug, pesticide or biocidal product use. |

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                             |            |
|---------------------------------------------|------------|
| Acute oral toxicity                         | Category 4 |
| Acute dermal toxicity                       | Category 4 |
| Acute Inhalation Toxicity - Dusts and Mists | Category 4 |

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Harmful if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician if you feel unwell

**Skin**

IF ON SKIN: Wash with plenty of soap and water  
Call a POISON CENTER or doctor/physician if you feel unwell  
Wash contaminated clothing before reuse

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Harmful to aquatic life with long lasting effects  
Contact with acids liberates very toxic gas

### 3. Composition/Information on Ingredients

| Component             | CAS No   | Weight % |
|-----------------------|----------|----------|
| Potassium thiocyanate | 333-20-0 | >95      |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Inhalation</b>                          | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Most important symptoms and effects</b> | . Metabolism may release cyanide, which may result in headache, dizziness, weakness, collapse, unconsciousness, and possible death: May cause cyanosis (bluish discoloration of skin due to deficient oxygenation of the blood)                                                                                                |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                          |

## 5. Fire-fighting measures

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Oxidizing Properties** Not oxidising

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Sulfur oxides. Potassium oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
2

**Flammability**  
1

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Avoid contact with skin and eyes. Keep people away from and upwind of spill/leak.

**Environmental Precautions** Do not flush into surface water or sanitary sewer system. Avoid release to the environment. Collect spillage. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from acids. Incompatible Materials. Strong oxidizing agents. Acids. Strong bases.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component             | ACGIH TLV | OSHA PEL                           | NIOSH                      | Mexico OEL (TWA) |
|-----------------------|-----------|------------------------------------|----------------------------|------------------|
| Potassium thiocyanate |           | (Vacated) TWA: 5 mg/m <sup>3</sup> | IDLH: 25 mg/m <sup>3</sup> |                  |

### Legend

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

|                                             |                                                                                                                                                                                                                                                   |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering Measures</b>                 | Ensure that eyewash stations and safety showers are close to the workstation location.<br>Ensure adequate ventilation, especially in confined areas.                                                                                              |
| <b><u>Personal Protective Equipment</u></b> |                                                                                                                                                                                                                                                   |
| <b>Eye/face Protection</b>                  | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.                                                                       |
| <b>Skin and body protection</b>             | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>               | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Recommended Filter type:</b>             | Particulates filter conforming to EN 143.                                                                                                                                                                                                         |
| <b>Hygiene Measures</b>                     | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## 9. Physical and chemical properties

|                                               |                               |
|-----------------------------------------------|-------------------------------|
| <b>Physical State</b>                         | Crystalline Solid             |
| <b>Appearance</b>                             | Colorless - White             |
| <b>Odor</b>                                   | Odorless                      |
| <b>Odor Threshold</b>                         | No information available      |
| <b>pH</b>                                     | 5.3-8.7 5% aq.solution        |
| <b>Melting Point/Range</b>                    | 170 - 179 °C / 338 - 354.2 °F |
| <b>Boiling Point/Range</b>                    | Decomposes                    |
| <b>Flash Point</b>                            | No information available      |
| <b>Evaporation Rate</b>                       | Not applicable                |
| <b>Flammability (solid,gas)</b>               | No information available      |
| <b>Flammability or explosive limits</b>       |                               |
| Upper                                         | No data available             |
| Lower                                         | No data available             |
| <b>Vapor Pressure</b>                         | <1 hPa @ 20 °C                |
| <b>Vapor Density</b>                          | Not applicable                |
| <b>Specific Gravity</b>                       | 1.886                         |
| <b>Bulk Density</b>                           | 750 - 1000 kg/m <sup>3</sup>  |
| <b>Solubility</b>                             | Soluble in water              |
| <b>Partition coefficient; n-octanol/water</b> | No data available             |
| <b>Autoignition Temperature</b>               | No information available      |
| <b>Decomposition Temperature</b>              | 500 °C                        |
| <b>Viscosity</b>                              | Not applicable                |
| <b>Molecular Formula</b>                      | C K N S                       |
| <b>Molecular Weight</b>                       | 97.18                         |

## 10. Stability and reactivity

|                                         |                                                                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | Yes                                                                                                                           |
| <b>Stability</b>                        | Light sensitive. Moisture sensitive. Air sensitive.                                                                           |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Avoid dust formation. Exposure to light. Exposure to moist air or water. Exposure to air. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Acids, Strong bases                                                                                  |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NOx), Sulfur oxides, Potassium oxides                                                                        |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                      |

**Hazardous Reactions** Contact with acids liberates very toxic gas.

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Component Information

| Component             | LD50 Oral                | LD50 Dermal               | LC50 Inhalation |
|-----------------------|--------------------------|---------------------------|-----------------|
| Potassium thiocyanate | LD50 = 854 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat ) | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component             | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------|----------|------------|------------|------------|------------|------------|
| Potassium thiocyanate | 333-20-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Metabolism may release cyanide, which may result in headache, dizziness, weakness, collapse, unconsciousness, and possible death: May cause cyanosis (bluish discoloration of skin due to deficient oxygenation of the blood)

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component             | Freshwater Algae | Freshwater Fish                           | Microtox   | Water Flea                        |
|-----------------------|------------------|-------------------------------------------|------------|-----------------------------------|
| Potassium thiocyanate | Not listed       | Oncorhynchus mykiss:<br>LC50: 11 mg/l/96h | Not listed | Daphnia Magna: EC50: 2.8 mg/l/96h |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

|                 |               |
|-----------------|---------------|
| <u>DOT</u>      | Not regulated |
| <u>TDG</u>      | Not regulated |
| <u>IATA</u>     | Not regulated |
| <u>IMDG/IMO</u> | Not regulated |

## 15. Regulatory information

**United States of America Inventory**

| Component             | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-----------------------|----------|------|-----------------------------------------------|-----------------------------|
| Potassium thiocyanate | 333-20-0 | X    | ACTIVE                                        | -                           |

**Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component             | CAS No   | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-----------------------|----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Potassium thiocyanate | 333-20-0 | X   | -    | 206-370-1 | X     | X    | X    | X    | X     | KE-29216 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component             | CAS No   | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|-----------------------|----------|----------|-------------------------------|---------------------------------|
| Potassium thiocyanate | 333-20-0 | >95      | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component             | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|-----------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Potassium thiocyanate | -                          | -                           | X                      | X                         |

**Clean Air Act**

| Component             | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|-----------------------|-----------|-------------------------|-------------------------|
| Potassium thiocyanate | X         |                         | -                       |

OSHA - Occupational Safety and Health Administration Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

California Proposition 65 This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component             | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-----------------------|---------------|------------|--------------|----------|--------------|
| Potassium thiocyanate | -             | X          | X            | X        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant Y  
DOT Severe Marine Pollutant N

U.S. Department of Homeland Security This product does not contain any DHS chemicals.

**Other International Regulations**

Mexico - Grade No information available

Authorisation/Restrictions according to EU REACH Not applicable

| Component             | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Potassium thiocyanate | 333-20-0 | -                                                                   | -                                                                             | -                                                                                                     |

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component             | CAS No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Potassium thiocyanate | 333-20-0 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

**Other International Regulations**

| Component | CAS No | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|--------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
|           |        |                                                                                           |                                                                                          |                            |                                    |

|                       |          |                |                |                |                |
|-----------------------|----------|----------------|----------------|----------------|----------------|
| Potassium thiocyanate | 333-20-0 | Not applicable | Not applicable | Not applicable | Not applicable |
|-----------------------|----------|----------------|----------------|----------------|----------------|

## 16. Other information

**Prepared By** Health, Safety and Environmental Department  
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www.thermofisher.com

**Creation Date** 15-Feb-2010  
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**Print Date** 29-Mar-2024  
**Revision Summary** New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**